

This is how Star Wars droid BB-8 may work, based on the patent

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BB-8 is already the best-known droid from the upcoming Star Wars: The Force Awakens, and Mark Hamill was one of the first actors to spill the news that the rolling droid was a practical effect. BB-8 later stole the show at Star Wars Celebration when he rolled across the stage, a moment you can watch below.

There have been a few hints at how BB-8 was brought to life, and some hobbyists have already begun to create their own home-brewed version of the character. We also know a toy is on the way. The new, and rather beautiful, site How Does BB-8 Work offers a few more educated guesses after looking at the patent.

"The first thing we have learnt from this patent is that there is a concept in robotics called holonomic motion," the site's text explains. "Ground-based holonomic robots are those who can instantaneously move in any direction on the horizontal plane. That makes them incredibly responsive. Both the Sphero and our little guy BB-8 are holonomic robots."

The site goes on to describe how the droid may work. "The main difference we find with the Sphero is the dynamic mast described by the patent," they continue. "There is also a control system that keeps everything in balance, making sure the mast is always in vertical position. This, if you think about it, makes the whole mechanism behave like a miniature Segway. The arm can rotate and, according to the patent, magnetically interact with an external element. That means BB-8's head is most likely controlled via magnetic interaction. Magnets at the end of the mast and roller magnets at the base of the head."

Head over there to watch two animations that show how the mechanism may work in practice. It's very clever, and as you can see above the system looks amazing in practice.

This is the power of a practical effect: Instead of everyone shrugging and saying a computer did it, hobbyists and enthusiasts are digging in to solve the mystery, and BB-8 is able to show up and "act" during live events. Once the toy is released, if it acts at all like the original prop, we'll know for sure, but this all makes sense so far.

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